Dimensional Shift Design Principle (Project D-Link): A Policy-Driven Architecture for Heterogeneous CPU–GPU–Quantum Computation

Abstract

This paper introduces the Dimensional Shift Design Principle, implemented in Project D-Link, a policy-driven computational architecture designed to unify heterogeneous execution substrates, including classical CPUs, GPUs (CUDA), and quantum backends (simulated and hardware QPUs). Computation is modeled as adaptive routing and transformation of dimensional states over a central Dimension Bus, enabling runtime reconfiguration, traceability, replayability, and hybrid quantum–classical workflows. The system is organized into semi-autonomous computational colonies communicating exclusively via the bus, which enforces loose coupling and fault isolation. A formal state-space model, policy optimization objective, stability argument, and information-theoretic interpretation are provided. The proposed framework represents a candidate post-pipeline paradigm for adaptive control, cyber-physical systems, and heterogeneous high-performance computing.

Index Terms

Heterogeneous computing, policy-driven architectures, dimensional computing, GPU acceleration, quantum–classical hybrid systems, adaptive computation.

I. Introduction

Modern computing systems increasingly combine heterogeneous processing substrates such as CPUs, GPUs, specialized accelerators, and emerging quantum processors. Traditional pipeline-based and static graph architectures exhibit limited adaptability under dynamic workloads, resource availability, and optimization objectives. This work proposes the Dimensional Shift Design Principle, which redefines computation as adaptive movement through a multidimensional state space governed by explicit policies.

II. Conceptual Foundations

Unlike static pipelines, computation is represented as traversal through state dimensions, each encoding semantic roles (e.g., detection, control, optimization), physical or logical execution

layers (CPU, GPU, QPU), and policy-scoring axes (error, latency, energy, information gain). This abstraction enables runtime reconfiguration without structural redesign.

III. System Architecture

The architecture is centered around the Dimension Bus, a routing and messaging fabric supporting metadata-rich dimensional states. Computational functionality is encapsulated in semi-autonomous colonies, including acquisition, detection, control, meta-control, and quantum-policy colonies. Colonies communicate exclusively via the bus, enabling modularity, scalability, and fault isolation.

IV. Formal Model

A dimensional state is defined as $D = (S, M, T, P, \Sigma)$, where S denotes data payload, M metadata, T transformation lineage, P policy constraints, and Σ a multi-objective scoring vector. The global state space is $D = S \times M \times T \times P \times R^k$. Policies optimize expected utility over state trajectories under resource constraints.

V. Stability and Information-Theoretic Interpretation

System dynamics induced by policy-driven routing satisfy contraction properties under bounded scoring functions, yielding stability in the sense of invariant attractor sets. From an information-theoretic perspective, each dimension acts as a communication channel, and routing decisions maximize mutual information subject to latency and energy constraints.

VI. GPU and Quantum Integration

GPU acceleration is realized through CUDA-based dimensional primitives. Quantum execution is integrated via a hybrid control loop, where quantum operations are modeled using density operators and measurement statistics are incorporated into classical policy evaluation.

VII. Significance and Applications

The Dimensional Shift framework constitutes a candidate post-pipeline computational paradigm applicable to adaptive control systems, cyber-physical architectures, heterogeneous high-performance computing, and hybrid quantum–classical research workflows.

VIII. Future Work

Future research directions include autonomous policy learning using reinforcement learning, formal verification of dimensional invariants and safety properties, FPGA-based hardware implementations of the Dimension Bus, large-scale distributed deployments, and deeper integration with near-term quantum hardware for probabilistic optimization and decision support.

Additional work will focus on benchmarking across diverse workloads and establishing standardized open-source interfaces.

IX. Conclusion

This preprint establishes the Dimensional Shift Design Principle as a unified architectural approach to heterogeneous computation, emphasizing adaptability, explainability, and extensibility across classical, GPU, and quantum substrates.

Appendix A: Czech Summary (Český souhrn)

Tato část je určena pro čtenáře českého jazyka.

Abstrakt— Tento článek představuje princip dimenzionálního posunu realizovaný v projektu D-Link jako politicky řízenou architekturu pro heterogenní výpočty zahrnující CPU, GPU a kvantové backendy. Výpočet je modelován jako adaptivní směrování dimenzionálních stavů po dimenzionální sběrnici s důrazem na trasovatelnost, opakovatelnost a hybridní kvantově-klasické workflow. Navržený přístup představuje post-pipeline paradigma vhodné pro adaptivní a kyber-fyzikální systémy.